

GREATER FUNCTIONALITY Demands Higher Levels Of TECHNOLOGY



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Companies are trying to maintain their competitive edge by adding more functionality to their products. This is making systems that were previously relatively simple more complex and difficult to design. Designers are being challenged to deliver greater functionality in the same space or, often, in a smaller space. This raises the stakes when designers choose a suitable hardware platform for their next project

What is not necessarily apparent from the outside is the level of complexity that goes into a design. Externally, two products may look very similar—and often do—but looking inside will reveal very different levels of functionality and technology. Electronic door locks are becoming popular, especially in hotels, but increasingly in residential applications as well. Earlier designs com-

prise a microcontroller (MCU) and a reader for the electronic keycard that may be based on reading a magnetic strip or using NFC.

Next-generation key locks are unlikely to have a key at all, and will rely on fingerprints or other biometrics such as face or voice. Naturally, delivering this will require far more internal sophistication.

Although, the lock is very different, the basics remain; the door is only opened if the correct credentials (in whatever form, that is) are presented. Physical size is likely to remain the same, especially to allow for easy and inexpensive retrofitting.

However, the new lock will need biometric sensors such as fingerprint reader or possibly a vision sensor/camera and, maybe, a microphone depending which biometrics (or a combination of biometrics) will form the basis of the lock. Each of the sensors will require some form of processing (such as high-speed digital signal processing, or DSP) to process the data for the system to grant access, or not.

In terms of software, designers can readily access significant resources to complete such a design. ARM core

